EQUAÇÃO DO MOMENTO HORIZONTAL

EQUAÇÃO DO MOMENTO HORIZONTAL

O ponto de partida são as componentes horizontais da Eq. (2.3), com (2.6):

$$\frac{D\vec{V}}{Dt} = \frac{1}{\rho_m}\nabla P - g\hat{k} - 2\Omega_e \times \vec{V} + \frac{\vec{F}_v}{\rho_d} \qquad (2.3)$$

$$-\left(2\Omega_e \times \vec{V}\right)_H = -fk \times \vec{V}_H - ew\hat{\imath}\,, \qquad -\left(2\Omega_e \times \vec{V}\right)_z = +eu \qquad (2.6)$$

$$\frac{D\vec{V}_H}{Dt} = -\frac{1}{\rho_m}\nabla_z P - f\hat{k} \times \vec{V}_H - ew\hat{\imath} + \frac{F}{\rho_d} \qquad (5.1)$$

$$\hat{k} \times \vec{V}_H = (-u, \quad v)$$

O trabalho consiste em reescrever cada termo em (x, y, ζ) e depois montá-los em forma de fluxo.

Advecção em ζ e identidade vetorial-invariante

$$\frac{D\varphi}{Dt} = \left.\frac{\partial\hat{\varphi}}{\partial t}\right|_{\zeta} + u\left.\frac{\partial\hat{\varphi}}{\partial x}\right|_{\zeta} + v\left.\frac{\partial\hat{\varphi}}{\partial y}\right|_{\zeta} + \omega\frac{\partial\hat{\varphi}}{\partial\zeta} \qquad (4.15)$$

Pela Eq. (4.15) aplicada a $\hat{\varphi} = \vec{V}_H$,

$$\frac{D\vec{V}_H}{Dt} = \frac{\partial\vec{V}_H}{\partial t} + (\vec{V}_H \cdot \nabla_\zeta)\vec{V}_H + \omega\frac{\partial\vec{V}_H}{\partial\zeta} \qquad (5.2)$$

O **termo central é a advecção horizontal** $(\vec{V}_H \cdot \nabla_\zeta)\vec{V}_H$ *ao longo da superfície* ζ.

Escrevendo as duas componentes e somando e subtraindo o mesmo termo em cada uma,

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = \left(u\frac{\partial u}{\partial x} + v\frac{\partial v}{\partial x}\right) - v\frac{\partial v}{\partial x} + v\frac{\partial u}{\partial y} = \left(\frac{\partial}{\partial x}\left(\frac{u^2}{2}\right) + \frac{\partial}{\partial x}\left(\frac{v^2}{2}\right)\right) - v\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) = \left(\frac{\partial}{\partial x}\left(\frac{u^2}{2} + \frac{v^2}{2}\right)\right) - v\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)$$

$$K = \left(\frac{u^2}{2} + \frac{v^2}{2}\right)$$

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = \left(u\frac{\partial u}{\partial x} + v\frac{\partial v}{\partial x}\right) - v\frac{\partial v}{\partial x} + v\frac{\partial u}{\partial y} = \left(\frac{\partial}{\partial x}\left(\frac{u^2}{2}\right) + \frac{\partial}{\partial x}\left(\frac{v^2}{2}\right)\right) - v\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) = \left(\frac{\partial K}{\partial x}\right) - v\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)$$

5.1 Passo 1 – Advecção em ζ e identidade vetorial-invariante

$$u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}=\left(u\frac{\partial u}{\partial y}+v\frac{\partial v}{\partial y}\right)+u\frac{\partial v}{\partial x}-u\frac{\partial u}{\partial y}=\left(\frac{\partial}{\partial y}\left(\frac{u^2}{2}\right)+\frac{\partial}{\partial y}\left(\frac{v^2}{2}\right)\right)+u\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right)=\left(\frac{\partial}{\partial y}\left(\frac{u^2}{2}+\frac{v^2}{2}\right)\right)+u\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right)$$

$$K=\left(\frac{u^2}{2}+\frac{v^2}{2}\right)$$

$$u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}=\left(u\frac{\partial u}{\partial y}+v\frac{\partial v}{\partial y}\right)+u\frac{\partial v}{\partial x}-u\frac{\partial u}{\partial y}=\left(\frac{\partial}{\partial y}\left(\frac{u^2}{2}\right)+\frac{\partial}{\partial y}\left(\frac{v^2}{2}\right)\right)+u\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right)=\frac{\partial K}{\partial y}+u\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right)$$

5.1 Passo 1 – Advecção em ζ e identidade vetorial-invariante

$$u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}=\left(u\frac{\partial u}{\partial x}+v\frac{\partial v}{\partial x}\right)-v\frac{\partial v}{\partial x}+v\frac{\partial u}{\partial y}=\left(\frac{\partial}{\partial x}\left(\frac{u^2}{2}\right)+\frac{\partial}{\partial x}\left(\frac{v^2}{2}\right)\right)-v\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right)=\left(\frac{\partial K}{\partial x}\right)-v\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right) \quad (5.3)$$

$$u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}=\left(u\frac{\partial u}{\partial y}+v\frac{\partial v}{\partial y}\right)+u\frac{\partial v}{\partial x}-u\frac{\partial u}{\partial y}=\left(\frac{\partial}{\partial y}\left(\frac{u^2}{2}\right)+\frac{\partial}{\partial y}\left(\frac{v^2}{2}\right)\right)+u\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right)=\left(\frac{\partial K}{\partial y}\right)+u\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right) \quad (5.4)$$

pois $\frac{\partial K}{\partial x}=\left(u\frac{\partial u}{\partial x}+v\frac{\partial v}{\partial x}\right)$ e $\frac{\partial K}{\partial y}=\left(u\frac{\partial u}{\partial y}+v\frac{\partial v}{\partial y}\right)$. Todas as derivadas são a ζ constante e $K\equiv\frac{u^2+v^2}{2}=\frac{1}{2}\left|\vec{V}_H\right|^2$.

Definindo a **vorticidade vertical relativa medida na superfície** ζ,

$$\zeta_r=\left.\frac{\partial v}{\partial x}\right|_\zeta-\left.\frac{\partial u}{\partial y}\right|_\zeta \quad (5.5)$$

as duas **componentes se reúnem na identidade**

$$\left(\vec{V}_H\cdot\nabla_\zeta\right)\vec{V}_H=(\nabla)_\zeta K+\zeta_r\hat{\boldsymbol{k}}\times\vec{V}_H \quad (\mathbf{5.6})$$

$$(\vec{V}_H \cdot \nabla_\zeta)\vec{V}_H = (\nabla)_\zeta K + \zeta_r \hat{\boldsymbol{k}} \times \vec{V}_H \qquad (\mathbf{5.6})$$

Esta identidade é *exata* em **ζ constante**, e é a ***versão bidimensional da identidade vetorial***

$$(\vec{V}_H \cdot \nabla)\vec{V}_H = \nabla K + (\nabla \times \vec{V}_H) \times \vec{V}_H.$$

Nenhum termo de inclinação foi desprezado: toda a influência do terreno está na troca de $\left.\frac{\partial}{\partial x}\right|_z$ por $\left.\frac{\partial}{\partial x}\right|_\zeta$ feita em (4.15).

$$\frac{D\varphi}{Dt} = \left.\frac{\partial \hat{\varphi}}{\partial t}\right|_z + u\left.\frac{\partial \hat{\varphi}}{\partial x}\right|_z + v\left.\frac{\partial \hat{\varphi}}{\partial y}\right|_z + w\frac{\partial \hat{\varphi}}{\partial z} \qquad (4.15)$$

$$\frac{D\varphi}{Dt} = \left.\frac{\partial \hat{\varphi}}{\partial t}\right|_\zeta + u\left.\frac{\partial \hat{\varphi}}{\partial x}\right|_\zeta + v\left.\frac{\partial \hat{\varphi}}{\partial y}\right|_\zeta + \omega\frac{\partial \hat{\varphi}}{\partial \zeta} \qquad (4.15)$$

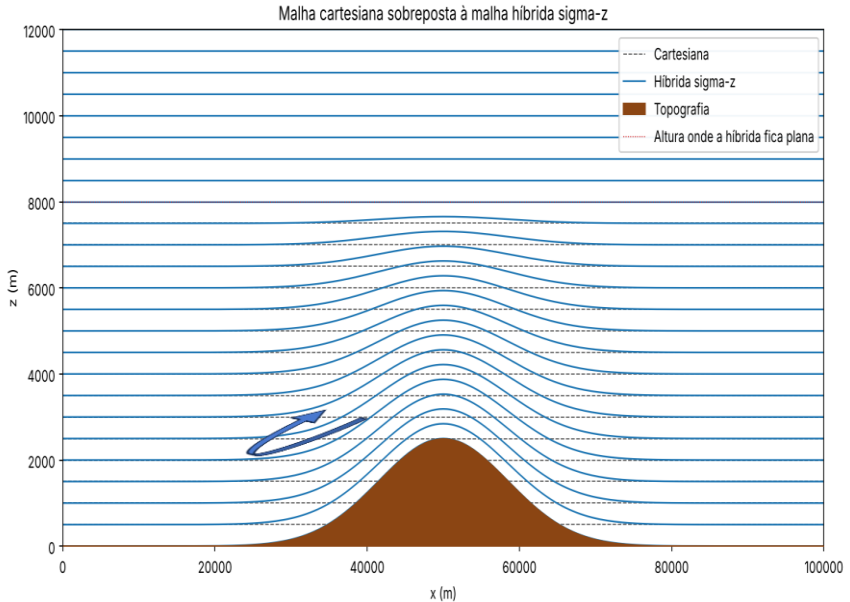
Significado de ζ_r. Pelo teorema de Stokes aplicado a um circuito fechado na superfície ζ, a circulação de $\vec{V}_H$ dividida pela área horizontal projetada é ζ_r. É isso que o MPAS calcula na malha, somando as velocidades normais nas arestas do triângulo da malha dual (SKAMAROCK et al., 2021).

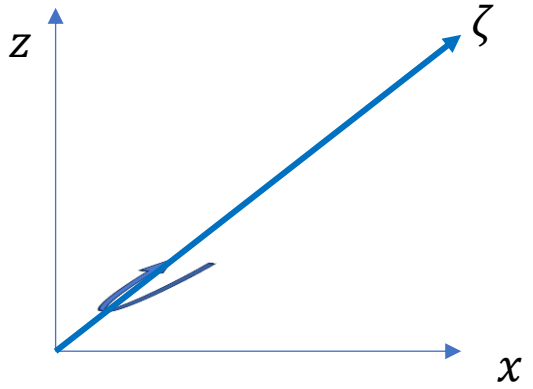
Resultado do passo 1.

$$\frac{D\vec{V}_H}{Dt} = \frac{\partial \vec{V}_H}{\partial t} + (\vec{V}_H \cdot \nabla_\zeta)\vec{V}_H + \omega \frac{\partial \vec{V}_H}{\partial \zeta} \qquad (5.2)$$

$$(\vec{V}_H \cdot \nabla_\zeta)\vec{V}_H = (\nabla)_\zeta K + \zeta_r \hat{\boldsymbol{k}} \times \vec{V}_H \qquad (\mathbf{5.6})$$

$$\frac{D\vec{V}_H}{Dt} = \frac{\partial \vec{V}_H}{\partial t} + \nabla_\zeta K + \zeta_r \hat{\boldsymbol{k}} \times \vec{V}_H + \omega \frac{\partial \vec{V}_H}{\partial \zeta} \qquad (5.7)$$





Gradiente de pressão em ζ

5.2 Passo 2 – Gradiente de pressão em ζ

A força de gradiente de pressão horizontal, a $z\ constante$, pela regra da cadeia (4.8), é

$$P(x,y,z) = \mathrm{P}(x,y,\zeta(x,y,z))$$

$$\left.\frac{\partial}{\partial x}\right|_z = \left.\frac{\partial}{\partial x}\right|_\zeta - \frac{z_x}{z_\zeta}\frac{\partial}{\partial \zeta}, \qquad \left.\frac{\partial}{\partial y}\right|_z = \left.\frac{\partial}{\partial y}\right|_\zeta - \frac{z_y}{z_\zeta}\frac{\partial}{\partial \zeta}, \qquad \frac{\partial}{\partial z} = \frac{1}{z_\zeta}\frac{\partial}{\partial \zeta} \tag{4.8}$$

$$\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)\Big|_z = \left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)\Big|_\zeta + \left(-\frac{z_x}{z_\zeta}-\frac{z_y}{z_\zeta}+\frac{1}{z_\zeta}\right)\frac{\partial}{\partial \zeta}$$

$$(\nabla)\Big|_z = (\nabla)\Big|_\zeta + \frac{1}{z_\zeta}(-z_x - z_y + 1)\frac{\partial}{\partial \zeta}$$

$$-z_H = -z_x - z_y + 1$$

$$z_H = z_x + z_y - 1$$

$$(\nabla)\Big|_z = (\nabla)\Big|_\zeta + \frac{1}{z_\zeta}(-z_H)\frac{\partial}{\partial \zeta}$$

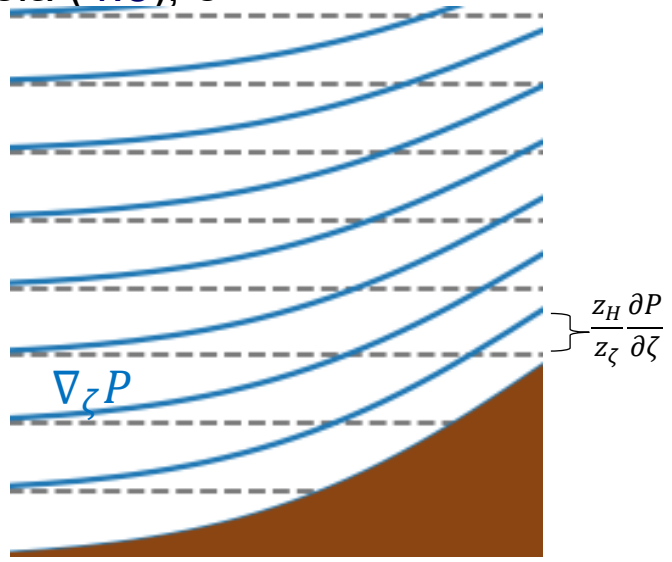
$$(\nabla)\Big|_z = (\nabla)\Big|_\zeta - \frac{z_H}{z_\zeta}\frac{\partial}{\partial \zeta}$$

5.2 Passo 2 – Gradiente de pressão em ζ

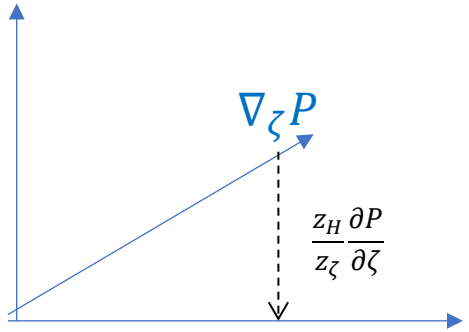
A **força de gradiente de pressão horizontal, a z *constante***, pela regra da cadeia (4.8), é

$$(\nabla)\big|_z = (\nabla)\big|_\zeta - \frac{z_H}{z_\zeta}\frac{\partial}{\partial\zeta}$$

$$-\frac{1}{\rho_m}\nabla_z P = -\frac{1}{\rho_m}\left[\nabla_\zeta P - \frac{z_H}{z_\zeta}\frac{\partial P}{\partial\zeta}\right] \qquad (5.8)$$



O primeiro termo é **a força medida ao longo da superfície** ζ e o **segundo a corrige de volta à horizontal**.



Será multiplicada por $\tilde{\rho}_d = \rho_d z_\zeta$:

$$-\frac{\rho_d z_\zeta}{\rho_m}\nabla_z P = -\frac{\rho_d z_\zeta}{\rho_m}\left[\nabla_\zeta P - \frac{z_H}{z_\zeta}\frac{\partial P}{\partial \zeta}\right] \qquad (5.8)$$

$$-\frac{\rho_d z_\zeta}{\rho_m}\nabla_z P = -\frac{\rho_d}{\rho_m}\left[z_\zeta \nabla_\zeta P - \frac{z_\zeta z_H}{z_\zeta}\frac{\partial P}{\partial \zeta}\right] \qquad (5.8)$$

$$-\frac{\tilde{\rho}_d}{\rho_m}\nabla_z P = -\frac{\rho_d}{\rho_m}\left[z_\zeta \nabla_\zeta P - z_H \frac{\partial P}{\partial \zeta}\right] \qquad (5.9)$$

Para *trazer isso à forma de fluxo* escreve-se o colchete como diferença de duas derivadas de produto. Note que se a identidade

$$\nabla_\zeta\left(z_\zeta P\right) = z_\zeta \nabla_\zeta (P) + P\nabla_\zeta\left(z_\zeta\right) \qquad (5.10)$$

$$\frac{\partial (z_H P)}{\partial \zeta} = z_H \frac{\partial (P)}{\partial \zeta} + P\frac{\partial (z_H)}{\partial \zeta} \qquad (5.11)$$

5.2 Passo 2 – Gradiente de pressão em ζ

$$\nabla_\zeta(z_\zeta P) = z_\zeta \nabla_\zeta(P) + P\nabla_\zeta(z_\zeta) \qquad (5.10)$$

$$\frac{\partial(z_H P)}{\partial\zeta} = z_H \frac{\partial(P)}{\partial\zeta} + P\frac{\partial(z_H)}{\partial\zeta} \qquad (5.11)$$

Note que

$$\nabla_\zeta\left(z_\zeta\right) = \frac{\partial}{\partial\zeta}(\nabla_\zeta z) = \frac{\partial}{\partial\zeta}(z_H)$$

(derivadas cruzadas de ***z***).

$$z_H = z_x + z_y - 1$$

Subtraindo as duas expressões

$$\nabla_\zeta(z_\zeta P) - \frac{\partial(z_H P)}{\partial\zeta} = z_\zeta \nabla_\zeta(P) + P\nabla_\zeta(z_\zeta) - z_H\frac{\partial(P)}{\partial\zeta} - P\frac{\partial(z_H)}{\partial\zeta}$$

$$\nabla_\zeta(z_\zeta P) - \frac{\partial(z_H P)}{\partial\zeta} = z_\zeta \nabla_\zeta(P) + P\frac{\partial}{\partial\zeta}(z_H) - z_H\frac{\partial(P)}{\partial\zeta} - P\frac{\partial(z_H)}{\partial\zeta}$$

$$\nabla_\zeta(z_\zeta P) - \frac{\partial(z_H P)}{\partial\zeta} = z_\zeta \nabla_\zeta(P) - z_H\frac{\partial(P)}{\partial\zeta}$$

$$\nabla_\zeta(z_\zeta P) - \frac{\partial(z_H P)}{\partial\zeta} = z_\zeta \nabla_\zeta(P) - z_H \frac{\partial(P)}{\partial\zeta}$$

Substituindo na 5.9

$$-\frac{\tilde{\rho}_d}{\rho_m}\nabla_z P = -\frac{\rho_d}{\rho_m}\left[z_\zeta \nabla_\zeta P - z_H \frac{\partial P}{\partial\zeta}\right] \qquad (5.9)$$

$$-\frac{\tilde{\rho}_d}{\rho_m}\nabla_z P = -\frac{\rho_d}{\rho_m}\left[\nabla_\zeta(z_\zeta P) - \frac{\partial(z_H P)}{\partial\zeta}\right] \qquad (5.9)$$

Usando $z_\zeta = 1/\zeta_z$ na última igualdade.

$$-\frac{\tilde{\rho}_d}{\rho_m}\nabla_z P = -\frac{\rho_d}{\rho_m}\left[\nabla_\zeta\left(\frac{P}{\zeta_z}\right) - \frac{\partial(z_H P)}{\partial\zeta}\right] \qquad (5.12)$$

$$-\frac{\tilde{\rho}_d}{\rho_m}\nabla_z P = -\frac{\rho_d}{\rho_m}\left[\nabla_\zeta\left(\frac{P}{\zeta_z}\right) - \frac{\partial(z_H P)}{\partial\zeta}\right] \qquad (5.12)$$

É a forma que aparece no slide do MPAS, com $\frac{\partial(z_H P)}{\partial\zeta}$ no segundo termo, é a Eq. (13) de Klemp (2011).

Na notação de Skamarock et al. (2021) escreve-se,

$$\zeta_H = \nabla_H \zeta\Big|_z \qquad z_H = -\frac{\zeta_H}{\zeta_z} \qquad \zeta_x = -\frac{z_x}{z_\zeta}$$

de modo que $\partial\zeta(\zeta Hp/\zeta z) = -\partial\zeta(zHp)$: é exatamente o mesmo

$$\frac{\partial}{\partial\zeta}\left(\frac{\zeta_H P}{\zeta_z}\right) = -\frac{\partial}{\partial\zeta}(z_H P)$$

5.2 Passo 2 – Gradiente de pressão em ζ

$$-\tilde{\rho}_d \frac{1}{\rho_m} \nabla_z p = -\frac{\rho_d}{\rho_m}\left[\nabla_\zeta\left(z_\zeta\, p\right) - \frac{\partial(z_H\, p)}{\partial\zeta}\right] = -\frac{\rho_d}{\rho_m}\left[\nabla_\zeta\left(\frac{p}{\zeta_z}\right) - \frac{\partial(z_H\, p)}{\partial\zeta}\right] \qquad (5.12)$$

$$-\frac{\tilde{\rho}_d}{\rho_m}\nabla_z P = -\frac{\rho_d}{\rho_m}\left[\nabla_\zeta\left(\frac{P}{\zeta_z}\right) - \frac{\partial(z_H P)}{\partial\zeta}\right] \qquad (5.12)$$

$$-\frac{\tilde{\rho}_d}{\rho_m}\nabla_z P = -\frac{\rho_d}{\rho_m}\left[\nabla_\zeta\left(z_\zeta P\right) - \frac{\partial(z_H P)}{\partial\zeta}\right] \qquad (5.9)$$

Por que a forma de fluxo. Numa atmosfera em repouso, $p = P(z)$ depende só da altura física.

Então $\left.\frac{\partial(P)}{\partial x}\right|_\zeta = P' z_x$ e $\frac{\partial(P)}{\partial\zeta} = P' z_\zeta$, e o colchete de (5.9) fica $z_\zeta P' z_x - z_x P' z_\zeta = 0$.

Cada um dos dois termos é grande sobre terreno inclinado, e a força é a pequena diferença entre eles, o que torna delicado o cálculo numérico.

A escolha de Klemp (2011) e de Klemp et al. (2003) para discretizar esses termos tem essa motivação.

Força de Coriolis

Paulo Yoshio Kubota

$$\frac{D\vec{V}_H}{Dt} = -\frac{1}{\rho_m}\nabla_z P - f\hat{k}\times\vec{V}_H - ew\hat{i} + \frac{F}{\rho_d} \qquad (5.1)$$

A **parte horizontal de Coriolis** em (5.1) contém dois termos: $-f\hat{k}\times\vec{V}_H - ew\hat{i}$

$$\frac{D\vec{V}_H}{Dt} = \frac{\partial\vec{V}_H}{\partial t} + \nabla_\zeta K + \zeta_r\hat{\boldsymbol{k}}\times\vec{V}_H + \omega\frac{\partial\vec{V}_H}{\partial\zeta} \qquad (5.7)$$

O **primeiro tem a mesma estrutura do termo de vorticidade** $\zeta_r\hat{\boldsymbol{k}}\times\vec{V}_H$ em (5.7), e vai se somar a ele no passo 5.

O segundo $-ew\hat{i}$ é o Coriolis associado a cos ϕ e fica separado.

Montagem em forma de fluxo

Substituindo (5.7) em (5.1) e multiplicando tudo por $\tilde{\rho}_d$:

$$\frac{D\vec{V}_H}{Dt} = -\frac{1}{\rho_m}\nabla_z P - f\hat{k}\times\vec{V}_H - ew\hat{i} + \frac{F}{\rho_d} \qquad (5.1)$$

$$\frac{D\vec{V}_H}{Dt} = \frac{\partial\vec{V}_H}{\partial t} + \nabla_\zeta K + \zeta_r\hat{\boldsymbol{k}}\times\vec{V}_H + \omega\frac{\partial\vec{V}_H}{\partial\zeta} \qquad (5.7)$$

$$\frac{\partial\vec{V}_H}{\partial t} + \nabla_\zeta K + \zeta_r\hat{\boldsymbol{k}}\times\vec{V}_H + \omega\frac{\partial\vec{V}_H}{\partial\zeta} = -\frac{1}{\rho_m}\nabla_z P - f\hat{k}\times\vec{V}_H - ew\hat{i} + \frac{F}{\rho_d} \qquad (5.1)$$

$$\frac{\partial\vec{V}_H}{\partial t} = -\nabla_\zeta K - \zeta_r\hat{\boldsymbol{k}}\times\vec{V}_H - \omega\frac{\partial\vec{V}_H}{\partial\zeta} - \frac{1}{\rho_m}\nabla_z P - f\hat{k}\times\vec{V}_H - ew\hat{i} + \frac{F}{\rho_d} \qquad (5.13)$$

$$\frac{\partial \vec{V}_H}{\partial t} = -\nabla_\zeta K - \zeta_r \hat{\boldsymbol{k}} \times \vec{V}_H - \omega \frac{\partial \vec{V}_H}{\partial \zeta} - \frac{1}{\rho_m} \nabla_z P - f\hat{k} \times \vec{V}_H - ew\hat{\imath} + \frac{F}{\rho_d} \qquad (5.13)$$

multiplicando tudo por $\tilde{\rho}_d$:

$$\tilde{\rho}_d \frac{\partial \vec{V}_H}{\partial t} = -\tilde{\rho}_d \nabla_\zeta K - \tilde{\rho}_d \zeta_r \hat{\boldsymbol{k}} \times \vec{V}_H - \tilde{\rho}_d \omega \frac{\partial \vec{V}_H}{\partial \zeta} - \frac{\tilde{\rho}_d}{\rho_m} \nabla_z P - \tilde{\rho}_d f\hat{k} \times \vec{V}_H - \tilde{\rho}_d ew\hat{\imath} + \tilde{\rho}_d \frac{F}{\rho_d} \qquad (5.13)$$

onde $F_{\vec{V}_H} = \frac{F}{\rho_d}$ Falta passar do lado esquerdo $\tilde{\rho}_d \frac{\partial \vec{V}_H}{\partial t}$ para a derivada do fluxo $\vec{V}_H = \tilde{\rho}_d \vec{V}_H$:

$$\frac{\partial (\tilde{\rho}_d \vec{V}_H)}{\partial t} = \tilde{\rho}_d \frac{\partial \vec{V}_H}{\partial t} + \vec{V}_H \frac{\partial \tilde{\rho}_d}{\partial t} \qquad (5.14)$$

O **termo extra $\vec{V}_H \frac{\partial \tilde{\rho}_d}{\partial t}$ é avaliado com a continuidade (4.24):**

$$\frac{\partial(\tilde{\rho}_d \vec{V}_H)}{\partial t} = \tilde{\rho}_d \frac{\partial \vec{V}_H}{\partial t} + \vec{V}_H \frac{\partial \tilde{\rho}_d}{\partial t} \qquad (5.14)$$

O termo extra $\vec{V}_H \frac{\partial \tilde{\rho}_d}{\partial t}$ é avaliado com a continuidade (4.24):

$$\frac{\partial \tilde{\rho}_d}{\partial t} = -\nabla_\zeta \cdot \vec{V}_H - \frac{\partial \Omega}{\partial \zeta}$$

$$\vec{V}_H \frac{\partial \tilde{\rho}_d}{\partial t} = -\vec{V}_H \nabla_\zeta \cdot \vec{V}_H - \vec{V}_H \frac{\partial \Omega}{\partial \zeta} \qquad (5.15)$$

Somando às duas equações e lembrando que $\tilde{\rho}_d \omega = \Omega$,

$$\frac{\partial(\tilde{\rho}_d \vec{V}_H)}{\partial t} = \tilde{\rho}_d \frac{\partial \vec{V}_H}{\partial t} - \vec{V}_H \nabla_\zeta \cdot \vec{V}_H - \vec{V}_H \frac{\partial \Omega}{\partial \zeta} \qquad (5.14)$$

$$\frac{\partial(\tilde{\rho}_d \vec{V}_H)}{\partial t} = \tilde{\rho}_d \frac{\partial \vec{V}_H}{\partial t} - \vec{V}_H \nabla_\zeta \cdot \vec{V}_H - \vec{V}_H \frac{\partial \Omega}{\partial \zeta} \qquad (5.14)$$

$$\tilde{\rho}_d \frac{\partial \vec{V}_H}{\partial t} = \frac{\partial(\tilde{\rho}_d \vec{V}_H)}{\partial t} + \vec{V}_H \nabla_\zeta \cdot \vec{V}_H + \vec{V}_H \frac{\partial \Omega}{\partial \zeta} \qquad (5.14)$$

lembrando que $\tilde{\rho}_d \omega = \Omega$,

$$\tilde{\rho}_d \frac{\partial \vec{V}_H}{\partial t} = -\tilde{\rho}_d \nabla_\zeta K - \tilde{\rho}_d \zeta_r \hat{\boldsymbol{k}} \times \vec{V}_H - \Omega \frac{\partial \vec{V}_H}{\partial \zeta} - \frac{\tilde{\rho}_d}{\rho_m} \nabla_z P - \tilde{\rho}_d f \hat{k} \times \vec{V}_H - \tilde{\rho}_d e w \hat{\imath} + \tilde{\rho}_d \frac{F}{\rho_d} \qquad (5.13)$$

$$\frac{\partial \tilde{\rho}_d \vec{V}_H}{\partial t} = -\tilde{\rho}_d \nabla_\zeta K - \tilde{\rho}_d (\zeta_r + f) \hat{\boldsymbol{k}} \times \vec{V}_H - \vec{V}_H \nabla_\zeta \cdot \vec{V}_H - \left[\Omega \frac{\partial \vec{V}_H}{\partial \zeta} + \vec{V}_H \frac{\partial \Omega}{\partial \zeta} \right] - \frac{\tilde{\rho}_d}{\rho_m} \nabla_z P - \tilde{\rho}_d e w \hat{\imath} + \tilde{\rho}_d \frac{F}{\rho_d} \qquad (5.16)$$

O colchete é uma derivada de produto:

$$\Omega \frac{\partial \vec{V}_H}{\partial \zeta} + \vec{V}_H \frac{\partial \Omega}{\partial \zeta} = \frac{\partial(\Omega \vec{V}_H)}{\partial \zeta} \qquad (5.17)$$

Vorticidade absoluta

$$\frac{\partial \tilde{\rho}_d \vec{V}_H}{\partial t} = -\tilde{\rho}_d \nabla_\zeta K - \tilde{\rho}_d (\zeta_r + f)\hat{\boldsymbol{k}} \times \vec{V}_H - \vec{V}_H \nabla_\zeta \cdot \vec{V}_H - \left[\Omega \frac{\partial \vec{V}_H}{\partial \zeta} + \vec{V}_H \frac{\partial \Omega}{\partial \zeta}\right] - \frac{\tilde{\rho}_d}{\rho_m} \nabla_z P - \tilde{\rho}_d e w \hat{\imath} + \tilde{\rho}_d \frac{F}{\rho_d} \qquad (5.16)$$

Os dois termos em $\hat{\boldsymbol{k}} \times \vec{V}_H$ de (5.16) somam-se na vorticidade vertical absoluta

$$\eta = \zeta_r + f \qquad \tilde{\rho}_d \eta \hat{\boldsymbol{k}} \times \vec{V}_H$$

$$\eta \hat{\boldsymbol{k}} \times \tilde{\rho}_d \vec{V}_H \qquad (5.18)$$

O fator η contém a vorticidade *relativa* ζ_r ,que depende do próprio campo de vento, e portanto o termo $-\eta \hat{\boldsymbol{k}} \times \tilde{\rho}_d \vec{V}_H$ é não linear. Ele contém o Coriolis tradicional (*f*) e uma parte da advecção (*ζr*).

É isso que o slide chama de “termo de Coriolis não linear”.

$$\frac{\partial \tilde{\rho}_d \vec{V}_H}{\partial t} = -\tilde{\rho}_d \nabla_\zeta K - \eta \hat{\boldsymbol{k}} \times \tilde{\rho}_d \vec{V}_H - \vec{V}_H \nabla_\zeta \cdot \vec{V}_H - \left[\Omega \frac{\partial \vec{V}_H}{\partial \zeta} + \vec{V}_H \frac{\partial \Omega}{\partial \zeta}\right] - \frac{\tilde{\rho}_d}{\rho_m} \nabla_z P - \tilde{\rho}_d e w \hat{\imath} + \tilde{\rho}_d \frac{F}{\rho_d} \qquad (5.16)$$

Reunindo (5.16), (5.18) e (5.12): $-\frac{\tilde{\rho}_d}{\rho_m}\nabla_z P = -\frac{\rho_d}{\rho_m}\left[\nabla_\zeta\left(\frac{P}{\zeta_z}\right) - \frac{\partial(z_H P)}{\partial\zeta}\right]$ (5.12)

$$\Omega\frac{\partial\vec{V}_H}{\partial\zeta} + \vec{V}_H\frac{\partial\Omega}{\partial\zeta} = \frac{\partial\left(\Omega\vec{V}_H\right)}{\partial\zeta} \qquad (5.17)$$

$$\frac{\partial\tilde{\rho}_d\vec{V}_H}{\partial t} = -\frac{\rho_d}{\rho_m}\left[\nabla_\zeta\left(\frac{P}{\zeta_z}\right) - \frac{\partial(z_H P)}{\partial\zeta}\right] - \tilde{\rho}_d\nabla_\zeta K - \vec{V}_H\nabla_\zeta\cdot\vec{V}_H - \left[\frac{\partial\left(\Omega\vec{V}_H\right)}{\partial\zeta}\right] - \eta\hat{\boldsymbol{k}}\times\tilde{\rho}_d\vec{V}_H - \tilde{\rho}_d ew\hat{\imath} + \tilde{\rho}_d\frac{F}{\rho_d} \qquad (5.16)$$

$$\frac{\partial\mathbf{V}_H}{\partial t} = \underbrace{-\frac{\rho_d}{\rho_m}\left[\nabla_\zeta\left(\frac{p}{\zeta_z}\right) - \frac{\partial(z_H\, p)}{\partial\zeta}\right] - \tilde{\rho}_d\nabla_\zeta K}_{\text{operador gradiente}} \underbrace{- \mathbf{v}_H\nabla_\zeta\cdot\mathbf{V}_H - \frac{\partial(\Omega\,\mathbf{v}_H)}{\partial\zeta}}_{\text{divergência de fluxo}} \underbrace{- \eta\,\mathbf{k}\times\mathbf{V}_H}_{\text{Coriolis não linear}} - \tilde{\rho}_d\, e\, w\,\mathbf{i} + \mathbf{F}V_H \qquad (5.19)$$

Esta é a Eq. (6) de Skamarock et al. (2021) na aproximação de plano tangente (sem os termos de curvatura)

MOMENTO VERTICAL, TERMODINÂMICA E SISTEMA FECHADO

Paulo Yoshio Kubota

6.1 Momento vertical

A segunda lei de Newton para uma parcela, num referencial que gira com a Terra com velocidade angular $\boldsymbol{\Omega}_e$,

A componente vertical da Eq. (2.3), com (2.6), é

$$\frac{D\vec{V}}{Dt} = -\frac{1}{\rho_m}\nabla P - g\hat{k} - 2\Omega_e \times \vec{V} + \frac{\vec{F}_v}{\rho_d} \qquad (2.3)$$

$$-\left(2\Omega_e \times \vec{V}\right)_H = -fk \times \vec{V}_H - ew\hat{\imath}\,, \qquad -\left(2\Omega_e \times \vec{V}\right)_z = +eu \qquad (2.6)$$

$\nabla p/\rho_m$ é a força de gradiente de pressão por unidade de massa,
$g\mathbf{k}$ a gravidade(constante em plano tangente; a variação com r entra no Capítulo 7) e
$2\Omega_e \times \vec{V}$ **é** a aceleração de Coriolis.
$g\hat{k}$ é A força centrífuga .
$\vec{F}_v$ é a força por unidade de volume devida ao atrito: contém o termo viscoso molecular $\rho\nu\nabla^2\vec{V}$(para ν constante) e, quando for o caso, os efeitos de subgrade.

Lema (forma de fluxo). Se a densidade ρ satisfaz a continuidade

$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) = 0$, então para qualquer ψ

$$\rho \frac{D\psi}{Dt} = \frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot (\rho \vec{V} \psi) \qquad (4.1)$$

$$\nabla \cdot (\rho\psi\vec{V}) = \rho\psi(\nabla \cdot \vec{V}\ ) + \vec{V} \cdot (\nabla \rho\psi)$$

Prova. **Expandindo as derivadas de produto do lado direito,**

$$\frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot (\rho \vec{V} \psi) = \rho \frac{\partial(\psi)}{\partial t} + \psi \frac{\partial(\rho)}{\partial t} + \rho\psi(\nabla \cdot \vec{V}\ ) + \vec{V} \cdot (\nabla \rho\psi)$$

$$\frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot (\rho \vec{V} \psi) = \rho \frac{\partial(\psi)}{\partial t} + \psi \frac{\partial(\rho)}{\partial t} + \rho\psi(\nabla \cdot \vec{V}\ ) + \vec{V} \cdot (\psi\nabla\rho + \rho\nabla\psi)$$

$$\frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot (\rho \vec{V} \psi) = \rho \frac{\partial(\psi)}{\partial t} + \psi \frac{\partial(\rho)}{\partial t} + \rho\psi(\nabla \cdot \vec{V}\ ) + (\psi\vec{V} \cdot \nabla\rho + \rho\vec{V} \cdot \nabla\psi)$$

$$\frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot (\rho \vec{V} \psi) = \rho \frac{\partial(\psi)}{\partial t} + \psi \frac{\partial(\rho)}{\partial t} + \psi(\rho\nabla \cdot \vec{V}\ ) + \psi\vec{V} \cdot \nabla\rho + (\rho\vec{V} \cdot \nabla\psi)$$

Lema (forma de fluxo). Se a densidade ρ satisfaz a continuidade

$$\nabla \cdot (\rho\psi\vec{V}) = \rho\psi(\nabla \cdot \vec{V}\ ) + \vec{V} \cdot (\nabla\rho\psi)$$

$$\frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot (\rho\vec{V}\psi) = \rho\frac{\partial(\psi)}{\partial t} + \psi\frac{\partial(\rho)}{\partial t} + \psi(\rho\nabla \cdot \vec{V}\ ) + \psi\vec{V} \cdot \nabla\rho + (\rho\vec{V} \cdot \nabla\psi)$$

$$\frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot (\rho\vec{V}\psi) = \rho\frac{\partial(\psi)}{\partial t} + \psi\frac{\partial(\rho)}{\partial t} + \psi(\rho\nabla \cdot \vec{V} + \vec{V} \cdot \nabla\rho) + (\rho\vec{V} \cdot \nabla\psi)$$

$$\frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot (\rho\vec{V}\psi) = \rho\frac{\partial(\psi)}{\partial t} + \psi\frac{\partial(\rho)}{\partial t} + \psi(\nabla \cdot (\rho\vec{V})\ ) + (\rho\vec{V} \cdot \nabla\psi)$$

$$\frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot (\rho\vec{V}\psi) = \rho\frac{\partial(\psi)}{\partial t} + \psi\frac{\partial(\rho)}{\partial t} + \psi\nabla \cdot (\rho\vec{V}) + \rho\vec{V} \cdot \nabla\psi$$

$$\frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot (\rho\vec{V}\psi) = \rho\frac{\partial(\psi)}{\partial t} + \rho\vec{V} \cdot \nabla\psi + \psi\left[\frac{\partial\rho}{\partial t} + \nabla \cdot (\rho\vec{V})\right] = \rho\frac{D\psi}{Dt}$$

$$\frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot (\rho\vec{V}\psi) = \rho\left[\frac{\partial(\psi)}{\partial t} + \vec{V} \cdot \nabla\psi\right] + \psi\left[\frac{\partial\rho}{\partial t} + \nabla \cdot (\rho\vec{V})\right] = \rho\frac{D\psi}{Dt}$$

Lema (forma de fluxo). Se a densidade ρ satisfaz a continuidade

$$\nabla \cdot (\rho\psi\vec{V}) = \rho\psi(\nabla \cdot \vec{V}\ ) + \vec{V} \cdot (\nabla\rho\psi)$$

$$\rho\frac{D\psi}{Dt} = \frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot (\rho\vec{V}\psi) \qquad (4.1)$$

$$\frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot (\rho\vec{V}\psi) = \rho\left[\frac{\partial(\psi)}{\partial t} + \vec{V} \cdot \nabla\psi\right] + \psi\left[\frac{\partial\rho}{\partial t} + \nabla \cdot (\rho\vec{V})\right] = \rho\frac{D\psi}{Dt}$$

$\frac{\partial\rho}{\partial t} + \nabla \cdot (\rho\vec{V}) = 0$, então para qualquer ψ

$$\frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot (\rho\vec{V}\psi) = \rho\left[\frac{D\psi}{Dt}\right] + \psi\left[\frac{\partial\rho}{\partial t} + \nabla \cdot (\rho\vec{V}) = 0\right] = \rho\frac{D\psi}{Dt}$$

pois o colchete é nulo pela continuidade

$$\frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot (\rho\vec{V}\psi) = \rho\left[\frac{D\psi}{Dt}\right] = \rho\frac{D\psi}{Dt}$$

$$\frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot (\rho\vec{V}\psi) = \rho\frac{D\psi}{Dt} \qquad (4.2)$$

$$\frac{Dw}{Dt} = -\frac{1}{\rho_m}\frac{\partial P}{\partial z} - g + eu + \frac{F_w}{\rho_d} \qquad (6.1)$$

Para a componente vertical o MPAS usa diretamente a forma de fluxo (Eq. (7) de Skamarock et al. (2021)).

$$\frac{\partial W}{\partial t} = -\frac{\rho_d}{\rho_m}\left[\frac{\partial P}{\partial \zeta} + g\left(\frac{r_0}{r}\right)^2 \tilde{\rho}_m\right] - \left(\nabla \cdot \vec{V}_H W\right)_\zeta + F_w + \beta_d \tilde{\rho}_d \left(\frac{\vec{V}_H \cdot \vec{V}_H}{r} + e\vec{V}_H \cdot \hat{\imath}\right)$$

Aplica-se o Lema (4.26) com $\psi = w$:

$$\frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot \left(\rho \vec{V} \psi\right) = \rho \frac{D\psi}{Dt} \qquad (4.2)$$

$$\tilde{\rho}_d \frac{D\psi}{Dt} = \frac{\partial \tilde{\rho}_d \psi}{\partial t} + \nabla_\zeta \cdot \left(\vec{V}_H \psi\right) + \frac{\partial(\Omega\psi)}{\partial \zeta} \qquad (4.26)$$

$$\tilde{\rho}_d \frac{Dw}{Dt} = \frac{\partial \tilde{\rho}_d w}{\partial t} + \nabla_\zeta \cdot \left(\vec{V}_H w\right) + \frac{\partial(\Omega w)}{\partial \zeta} \qquad (4.26)$$

$$\rho \frac{D\psi}{Dt} = \frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot (\rho \vec{V} \psi) \qquad (4.2)$$

$$\boxed{\Omega \equiv \tilde{\rho}_d \omega} \qquad (4.22)$$

$$\tilde{\rho}_d \frac{Dw}{Dt} = \frac{\partial \tilde{\rho}_d w}{\partial t} + \nabla_\zeta \cdot (\vec{V}_H w) + \frac{\partial(\Omega w)}{\partial \zeta} \qquad (4.26)$$

$$\tilde{\rho}_d \frac{Dw}{Dt} = \frac{\partial W}{\partial t} + \nabla_\zeta \cdot (\vec{V}_H w) + \frac{\partial(\Omega w)}{\partial \zeta} \qquad (4.26)$$

$$\tilde{\rho}_d \frac{Dw}{Dt} = \frac{\partial W}{\partial t} + (\nabla \cdot \overrightarrow{V_H} w)_\zeta + \frac{\partial(\Omega w)}{\partial \zeta} \qquad (4.26)$$

$$\vec{V} = V(u, v, \Omega)$$

$$\tilde{\rho}_d \frac{Dw}{Dt} = \frac{\partial W}{\partial t} + (\nabla \cdot \vec{V} w)_\zeta \qquad (6.2)$$

$$\frac{Dw}{Dt} = -\frac{1}{\rho_m}\frac{\partial P}{\partial z} - g + eu + \frac{F_w}{\rho_d} \qquad (6.1)$$

Multiplicando (6.1) por $\tilde{\rho}_d$ e usando $\frac{\partial}{\partial z} = \frac{1}{z_\zeta}\frac{\partial}{\partial \zeta}$ de (4.8), o gradiente vertical de pressão fica

$$\tilde{\rho}_d \frac{Dw}{Dt} = -\frac{\tilde{\rho}_d}{\rho_m}\frac{\partial P}{\partial z} - \tilde{\rho}_d g + \tilde{\rho}_d eu + \frac{\tilde{\rho}_d F_w}{\rho_d} \qquad (6.1)$$

$$\left.\frac{\partial}{\partial x}\right|_z = \left.\frac{\partial}{\partial x}\right|_\zeta - \frac{z_x}{z_\zeta}\frac{\partial}{\partial \zeta}, \qquad \left.\frac{\partial}{\partial y}\right|_z = \left.\frac{\partial}{\partial y}\right|_\zeta - \frac{z_y}{z_\zeta}\frac{\partial}{\partial \zeta}, \qquad \frac{\partial}{\partial z} = \frac{1}{z_\zeta}\frac{\partial}{\partial \zeta} \qquad (4.8)$$

$$\tilde{\rho}_d \frac{Dw}{Dt} = -\frac{\tilde{\rho}_d}{\rho_m}\frac{1}{z_\zeta}\frac{\partial P}{\partial \zeta} - \tilde{\rho}_d g + \tilde{\rho}_d eu + \frac{\tilde{\rho}_d F_w}{\rho_d} \qquad (6.3)$$

$$\tilde{\rho}_d = z_\zeta \rho_d$$

$$\tilde{\rho}_d \frac{Dw}{Dt} = -\frac{\rho_d}{\rho_m}\frac{\partial P}{\partial \zeta} - \tilde{\rho}_d g + \tilde{\rho}_d eu + \frac{\tilde{\rho}_d F_w}{\rho_d} \qquad (6.3)$$

$$\tilde{\rho}_d \frac{Dw}{Dt} = -\frac{\rho_d}{\rho_m}\frac{\partial P}{\partial \zeta} - \tilde{\rho}_d g + \tilde{\rho}_d eu + \frac{\tilde{\rho}_d F_w}{\rho_d} \qquad (6.3)$$

sem nenhum termo de inclinação (a derivada vertical em z e em ζ coincidem a menos do fator $z\zeta$). Para a gravidade, define-se a densidade total acoplada

$$\tilde{\rho}_m \equiv \tilde{\rho}_d \frac{\rho_m}{\rho_d} = \tilde{\rho}_d \left(1 + \sum_j q_j\right) \qquad (6.4)$$

$$\tilde{\rho}_d \equiv \tilde{\rho}_m \frac{\rho_d}{\rho_m}$$

de modo que $\tilde{\rho}_d g = \frac{\rho_d}{\rho_m} g \tilde{\rho}_m$. Reunindo os termos

$$\tilde{\rho}_d \frac{Dw}{Dt} = -\frac{\rho_d}{\rho_m}\frac{\partial P}{\partial \zeta} - \frac{\rho_d}{\rho_m} g \tilde{\rho}_m + \tilde{\rho}_d eu + \frac{\tilde{\rho}_d F_w}{\rho_d} \qquad (6.3)$$

$$\tilde{\rho}_d \frac{Dw}{Dt} = -\frac{\rho_d}{\rho_m}\left[\frac{\partial P}{\partial \zeta} + g\tilde{\rho}_m\right] + \tilde{\rho}_d eu + \frac{\tilde{\rho}_d F_w}{\rho_d} \qquad (6.3)$$

$$\tilde{\rho}_d \frac{Dw}{Dt} = -\frac{\rho_d}{\rho_m}\left[\frac{\partial P}{\partial \zeta} + g\tilde{\rho}_m\right] + \tilde{\rho}_d eu + \frac{\tilde{\rho}_d F_w}{\rho_d} \qquad (6.3)$$

$$\tilde{\rho}_d \frac{Dw}{Dt} = \frac{\partial W}{\partial t} + \left(\nabla \cdot \vec{V} w\right)_\zeta \qquad (6.2)$$

$$\frac{\partial W}{\partial t} + \left(\nabla \cdot \vec{V} w\right)_\zeta = -\frac{\rho_d}{\rho_m}\left[\frac{\partial P}{\partial \zeta} + g\tilde{\rho}_m\right] + \tilde{\rho}_d eu + \frac{\tilde{\rho}_d F_w}{\rho_d} \qquad (6.3)$$

$$\frac{\partial W}{\partial t} = -\frac{\rho_d}{\rho_m}\left[\frac{\partial P}{\partial \zeta} + g\tilde{\rho}_m\right] - \left(\nabla \cdot \vec{V} w\right)_\zeta + \tilde{\rho}_d eu + \frac{\tilde{\rho}_d F_w}{\rho_d} \qquad (6.5)$$

Esta é a Eq. (7) de Skamarock et al. (2021) no plano tangente. O colchete mostra o equilíbrio hidrostático: com $w = 0$ e sem termos de fonte, $\left[\frac{\partial P}{\partial \zeta} + g\tilde{\rho}_m\right]$, que é a equação hidrostática em coordenada ζ. O empuxo de Boussinesq do primeiro slide aparece aqui completo, sem separação em estado básico mais perturbação.

Temperatura potencial úmida acoplada

Do Lema (4.26) com $\psi = \theta_m$ e da Eq. (2.26),

$$\boxed{\Omega \equiv \tilde{\rho}_d \omega}$$

$$\tilde{\rho}_d \frac{D\psi}{Dt} = \frac{\partial \tilde{\rho}_d \psi}{\partial t} + \nabla_\zeta \cdot \left(\vec{V}_H \psi\right) + \frac{\partial (\Omega \psi)}{\partial \zeta} \qquad (4.26)$$

$$\tilde{\rho}_d \frac{D\theta_m}{Dt} = \frac{\partial \tilde{\rho}_d \theta_m}{\partial t} + \nabla_\zeta \cdot \left(\vec{V}_H \theta_m\right) + \frac{\partial (\Omega \theta_m)}{\partial \zeta} \qquad (4.26)$$

$$\tilde{\rho}_d \frac{D\theta_m}{Dt} = \frac{\partial \tilde{\rho}_d \theta_m}{\partial t} + \nabla_\zeta \cdot \left(\vec{V}_H \theta_m\right) + \frac{\partial (\Omega \theta_m)}{\partial \zeta} \qquad (4.26)$$

$$\vec{V} = V(u, v, \Omega)$$

$$\tilde{\rho}_d \frac{D\theta_m}{Dt} = \frac{\partial \tilde{\rho}_d \theta_m}{\partial t} + \left(\nabla \cdot \left(\vec{V} \theta_m\right)\right)_\zeta \qquad (4.26)$$

$$\frac{D\theta_m}{Dt} = H_m \qquad (2.26)$$

Do Lema (4.26) com $\psi = \theta m$ e da Eq. (2.26),

$$\boxed{\Omega \equiv \tilde{\rho}_d \omega}$$

$$\frac{D\theta_m}{Dt} = H_m \qquad (2.26)$$

$$\tilde{\rho}_d \frac{D\theta_m}{Dt} = \tilde{\rho}_d H_m \qquad (2.26)$$

$$\frac{\partial \tilde{\rho}_d \theta_m}{\partial t} + \left(\nabla \cdot \left(\vec{V}\theta_m\right)\right)_\zeta = \tilde{\rho}_d H_m \qquad (6.6)$$

$$\frac{\partial \Theta}{\partial t} + \left(\nabla \cdot \left(\vec{V}\theta_m\right)\right)_\zeta = F_\Theta \qquad (6.7)$$

Aqui F_Θ reúne o aquecimento diabático (radiação, condensação, difusão turbulenta de calor etc.) e é o destino dos termos $J/Cp\ e\ -\partial w'T'/\partial z$ do slide de partida.

Escalares

Do mesmo modo, com $\psi = qj$ e (2.11),

$$\frac{\partial Q_j}{\partial t} + \left(\nabla \cdot \left(\vec{V} q_j\right)\right)_\zeta = F_{Q_j} \qquad (6.8)$$

Para o escalar genérico $\bar{c}$ do slide de partida, a equação é a mesma com $qj \rightarrow \bar{c}$.

Pressão: variável diagnóstica

A pressão que aparece em (5.19) e (6.5) não é prognosticada.

Ela sai da equação de estado (2.25). Como $\Theta_m = \tilde{\rho}_d \theta_m = \rho_d \, z_\zeta \theta_m$, *vale* $\rho_d \theta_m = \frac{\Theta_m}{z_\zeta}$

$$p = p_0 \left(\frac{R_d \, \Theta_m}{p_0 \, z_\zeta} \right)^\gamma . \qquad (6.9)$$

Esse é o ponto em que o modelo *fecha*: *p* depende de Θ_m e do jacobiano, e retroalimenta $\boldsymbol{V_H}$ e W pelos gradientes de pressão.

6.5 O sistema completo

Reunindo os resultados dos Capítulos 4 e 5 e deste capítulo, o sistema prognóstico do solver não hidrostático do MPAS, no plano tangente e com $\mathbf{V} = (\mathbf{V}_H, \Omega)$, é:

$$\frac{\partial \mathbf{V}_H}{\partial t} = -\frac{\rho_d}{\rho_m}\left[\nabla_\zeta\left(\frac{p}{\zeta_z}\right) - \frac{\partial (z_H p)}{\partial \zeta}\right] - \tilde{\rho}_d \nabla_\zeta K$$
$$- \eta\, \mathbf{k} \times \mathbf{V}_H - \mathbf{v}_H \nabla_\zeta \cdot \mathbf{V}_H - \frac{\partial (\Omega \mathbf{v}_H)}{\partial \zeta} - \tilde{\rho}_d e\, w\, \mathbf{i} + \mathbf{F}_{V_H}, \qquad \text{(M1)}$$

$$\frac{\partial W}{\partial t} = -\frac{\rho_d}{\rho_m}\left[\frac{\partial p}{\partial \zeta} + g\tilde{\rho}_m\right] - (\nabla \cdot \mathbf{V} w)_\zeta + \tilde{\rho}_d\, e\, u + F_W, \qquad \text{(M2)}$$

$$\frac{\partial \Theta_m}{\partial t} = -(\nabla \cdot \mathbf{V} \theta_m)_\zeta + F_{\Theta_m}, \qquad \text{(M3)}$$

$$\frac{\partial \tilde{\rho}_d}{\partial t} = -(\nabla \cdot \mathbf{V})_\zeta, \qquad \text{(M4)}$$

$$\frac{\partial Q_j}{\partial t} = -(\nabla \cdot \mathbf{V} q_j)_\zeta + F_{Q_j}, \qquad \text{(M5)}$$